

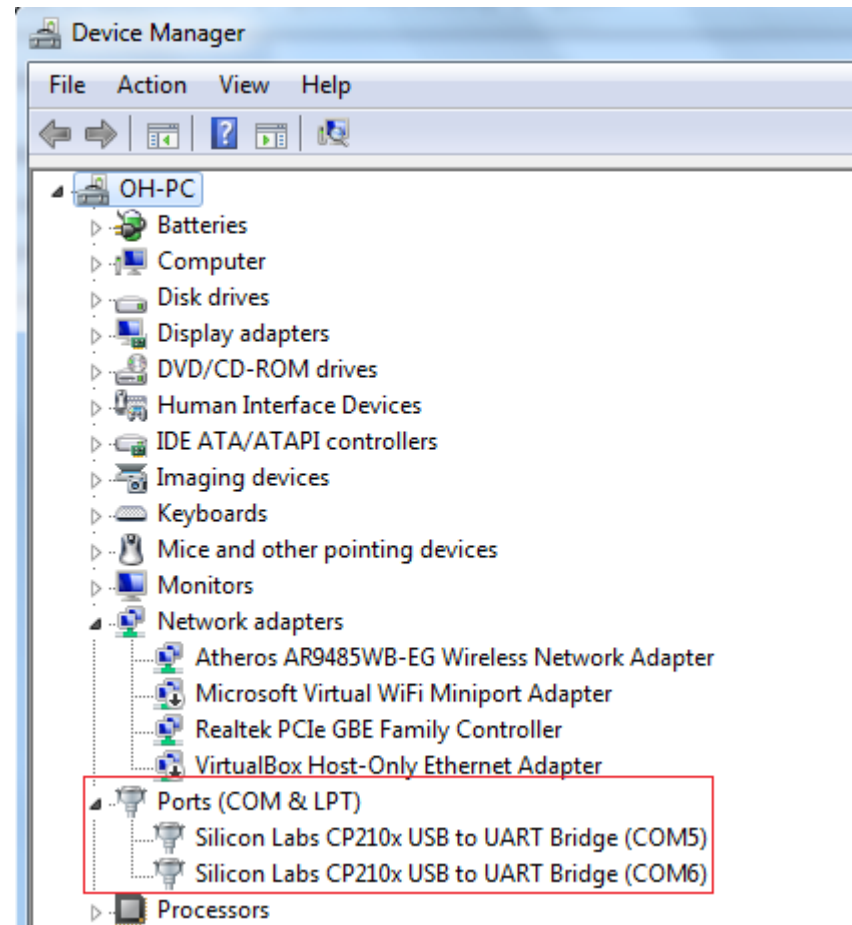
Getting Started with PX1122R RTK Evaluation Board

Install USB Driver from Silicon Lab Site

<https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>

After connecting both two USB cables, user should see 2 COM ports appear on Device Manager.

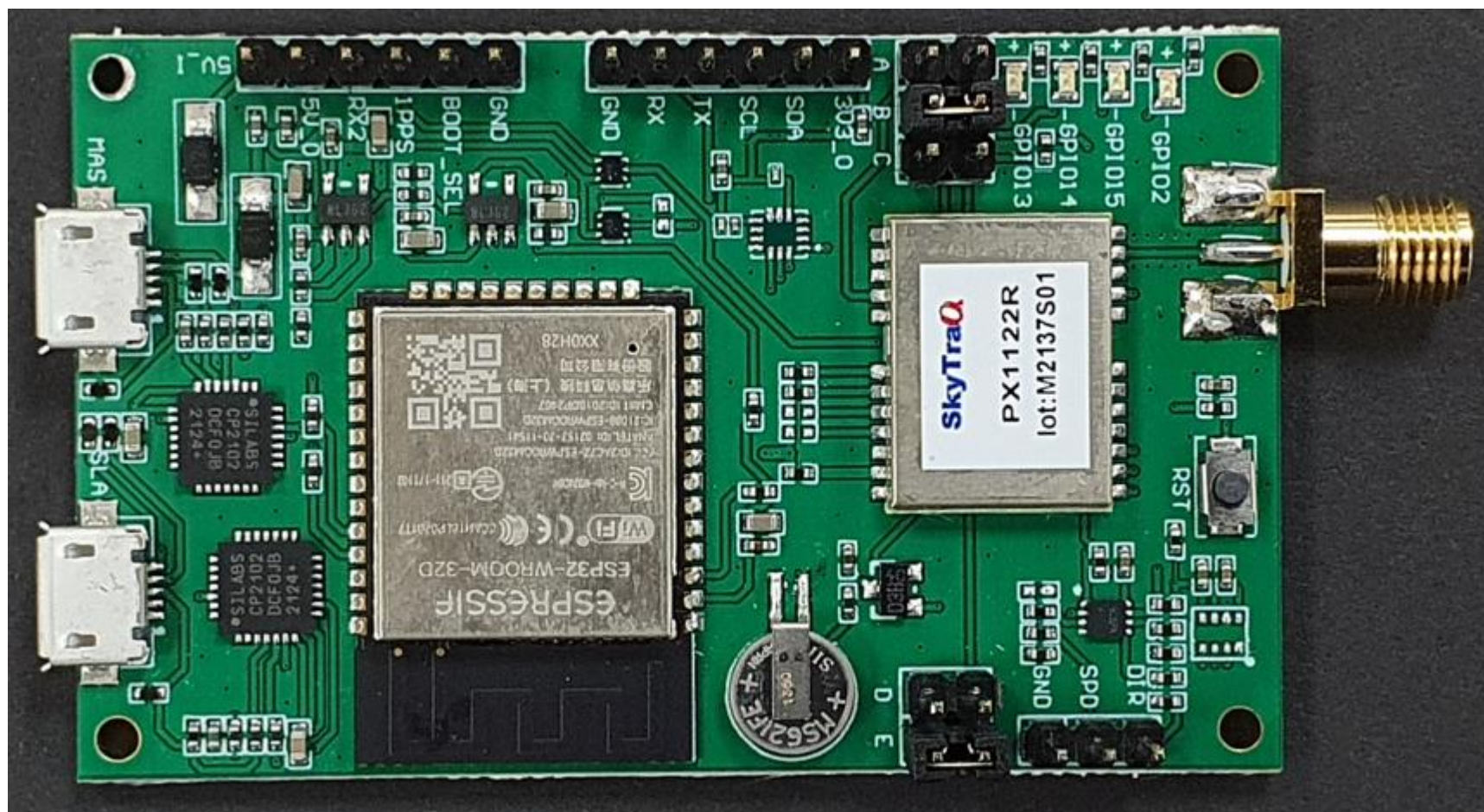
Please take note of which physical connection results in which COM port



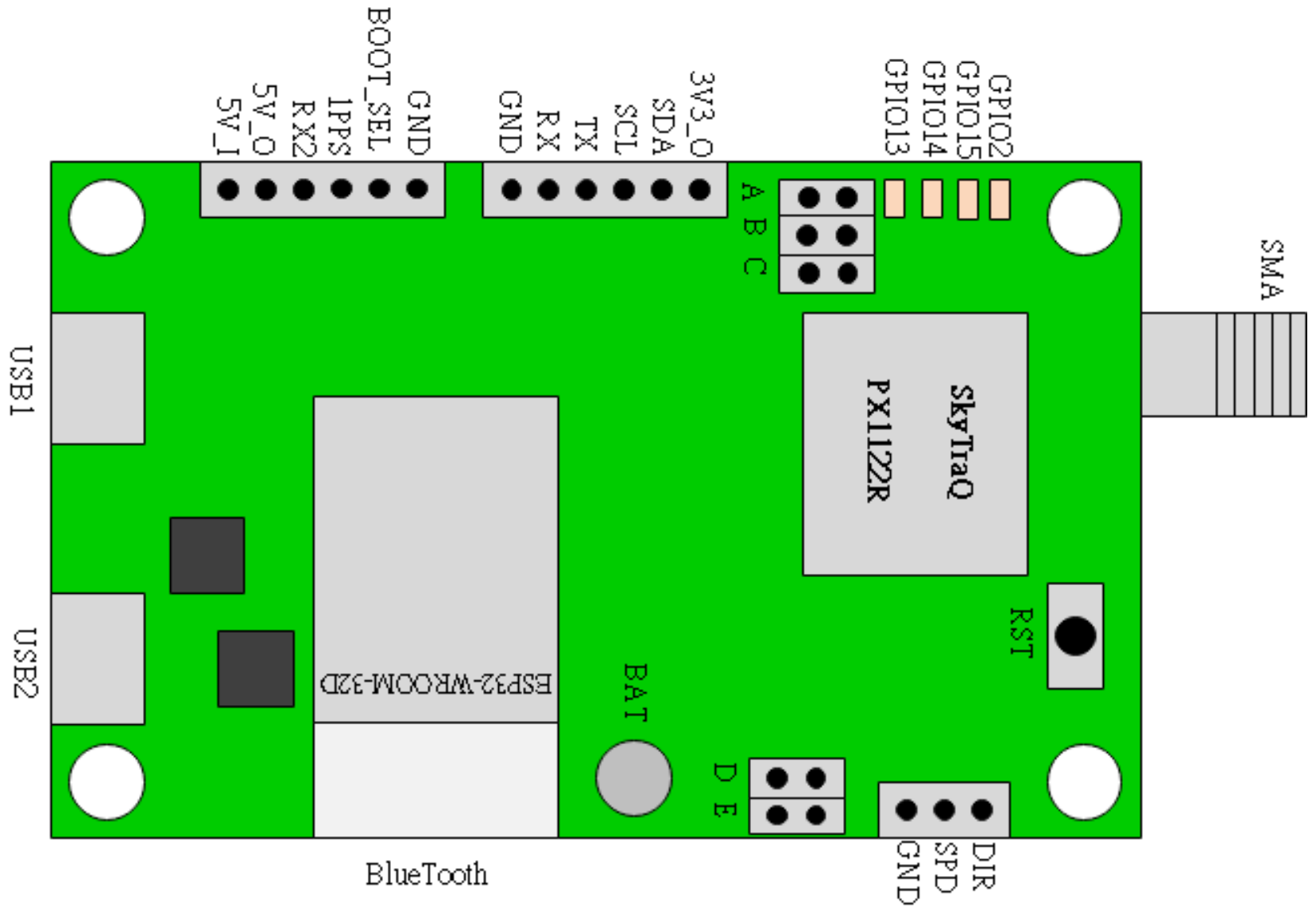
Get Windows GNSS Viewer

- https://navspark.mybigcommerce.com/content/GNSS_Viewer-CustomerRelease.zip
- <http://navspark.mybigcommerce.com/content/GNSSViewerUserGuide.pdf>

Top View of RTK EVB



Interface of RTK EVB (1/4)



Interface of RTK EVB (2/4)

NAME	DESCRIPTION
USB1	USB port used for NMEA output and configuration with GNSS Viewer
USB2	USB port used for receiving RTK correction data from base
Bluetooth	The Bluetooth supports SPP protocol to receive RTK correction data from Android phone running NTRIP client, and sending NMEA output to Android phone. Using Mock Location setting on Android, GIS data collector App can use external RTK receiver result.
SMA	The RF connector for GNSS antenna
5V_I	An extra 5V power input
5V_O	5V power output from USB1 or 5V_I
RX2	Extra UART Rx pin for receiving RTK correction data from base, see jumper A
1PPS	Pin for generating an 1Hz pulse with pulse width is about 100ms
BOOT_SEL	Pin for choosing which mode to boot, given 'H' during boot time selects flash mode and 'L' selects ROM mode. This pin is pulled high internally.
RX	Input pin for alternative UART Rx data to replace USB1, see jumper D
TX	Output pin for UART Tx NMEA message output

Interface of RTK EVB (3/4)

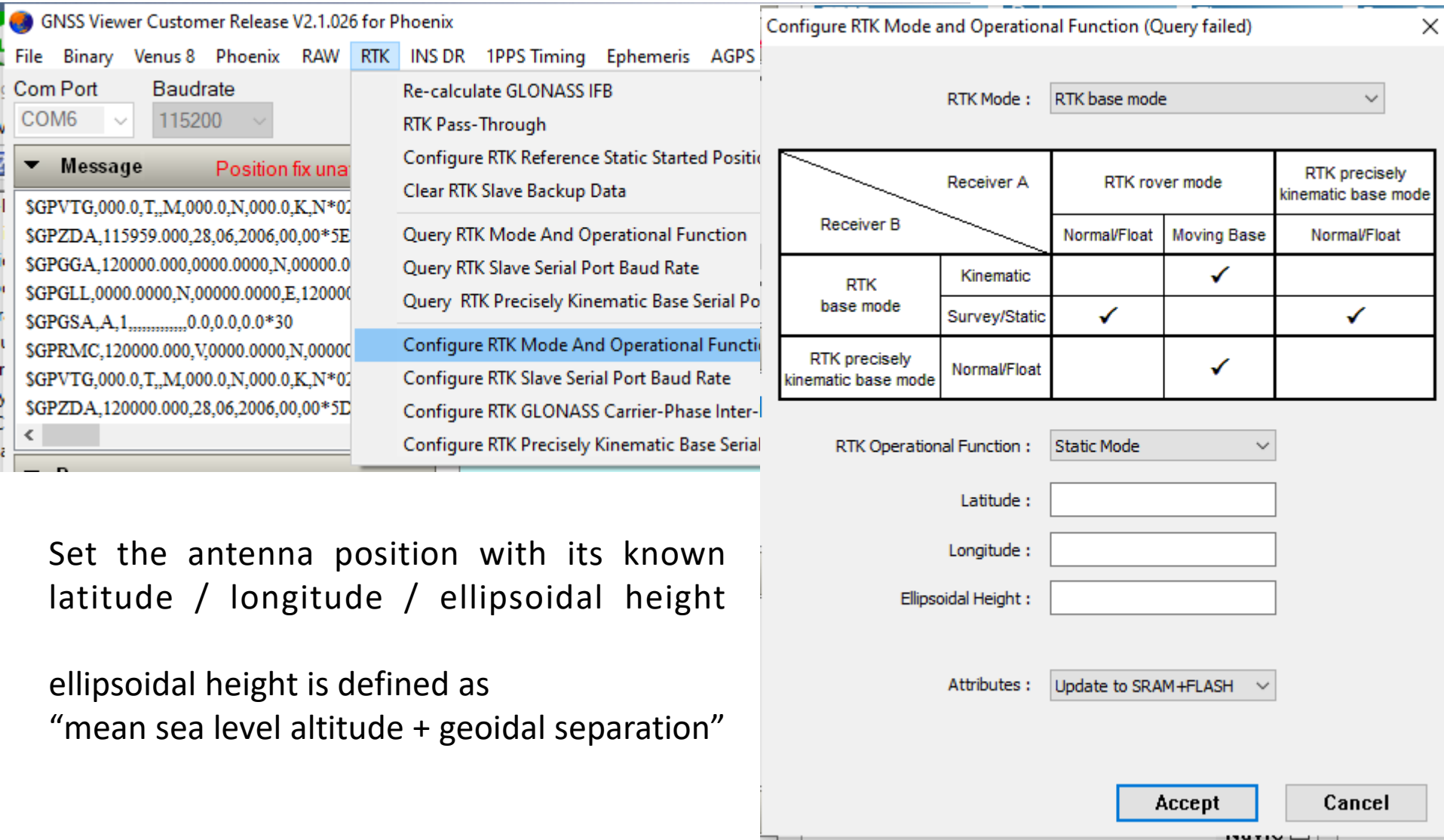
NAME	DESCRIPTION
SDA	SDA pin of I ² C interface for external slave device
SCL	SCL pin of I ² C interface for external slave device
3V3_O	3.3V power output from on-board LDO
GPIO15	LED indicator that can be used by module firmware
GPIO14	LED indicator that can be used by module firmware
GPIO13	LED indicator that can be used by module firmware
GPIO2	LED indicator that can be used by module firmware
RST	Reset button
DIR	Reserved pin
SPD	Reserved pin
GND	Ground pin

Interface of RTK EVB (4/4)

NAME	DESCRIPTION
A/B/C	Jumpers to select the source of RTK correction data to PX1122R pin-6 RXD2 A : from connector pin RX2 B : from USB2 C : from Bluetooth
D/E	Jumpers to select the source of PX1122R pin-21 RXD D : from connector pin RX E : from USB1

Setup EVB as RTK Base (1/3)

Setup RTK base position: if base antenna position is known



The screenshot shows the GNSS Viewer Customer Release V2.1.026 for Phoenix interface. The 'RTK' menu is open, displaying options such as 'Re-calculate GLONASS IFB', 'RTK Pass-Through', 'Configure RTK Reference Static Started Position', 'Clear RTK Slave Backup Data', 'Query RTK Mode And Operational Function', 'Query RTK Slave Serial Port Baud Rate', 'Query RTK Precisely Kinematic Base Serial Port Baud Rate', 'Configure RTK Mode And Operational Function', 'Configure RTK Slave Serial Port Baud Rate', 'Configure RTK GLONASS Carrier-Phase Interpolation', and 'Configure RTK Precisely Kinematic Base Serial Port Baud Rate'. The 'Configure RTK Mode And Operational Function' option is highlighted.

The 'Configure RTK Mode and Operational Function (Query failed)' dialog box is open, showing the following settings:

- RTK Mode : RTK base mode
- RTK Operational Function : Static Mode
- Latitude : [Empty field]
- Longitude : [Empty field]
- Ellipsoidal Height : [Empty field]
- Attributes : Update to SRAM+FLASH

The dialog box also contains a table for configuring RTK modes and operational functions:

Receiver A \ Receiver B		RTK rover mode		RTK precisely kinematic base mode
		Normal/Float	Moving Base	Normal/Float
RTK base mode	Kinematic		✓	
	Survey/Static	✓		✓
RTK precisely kinematic base mode	Normal/Float		✓	

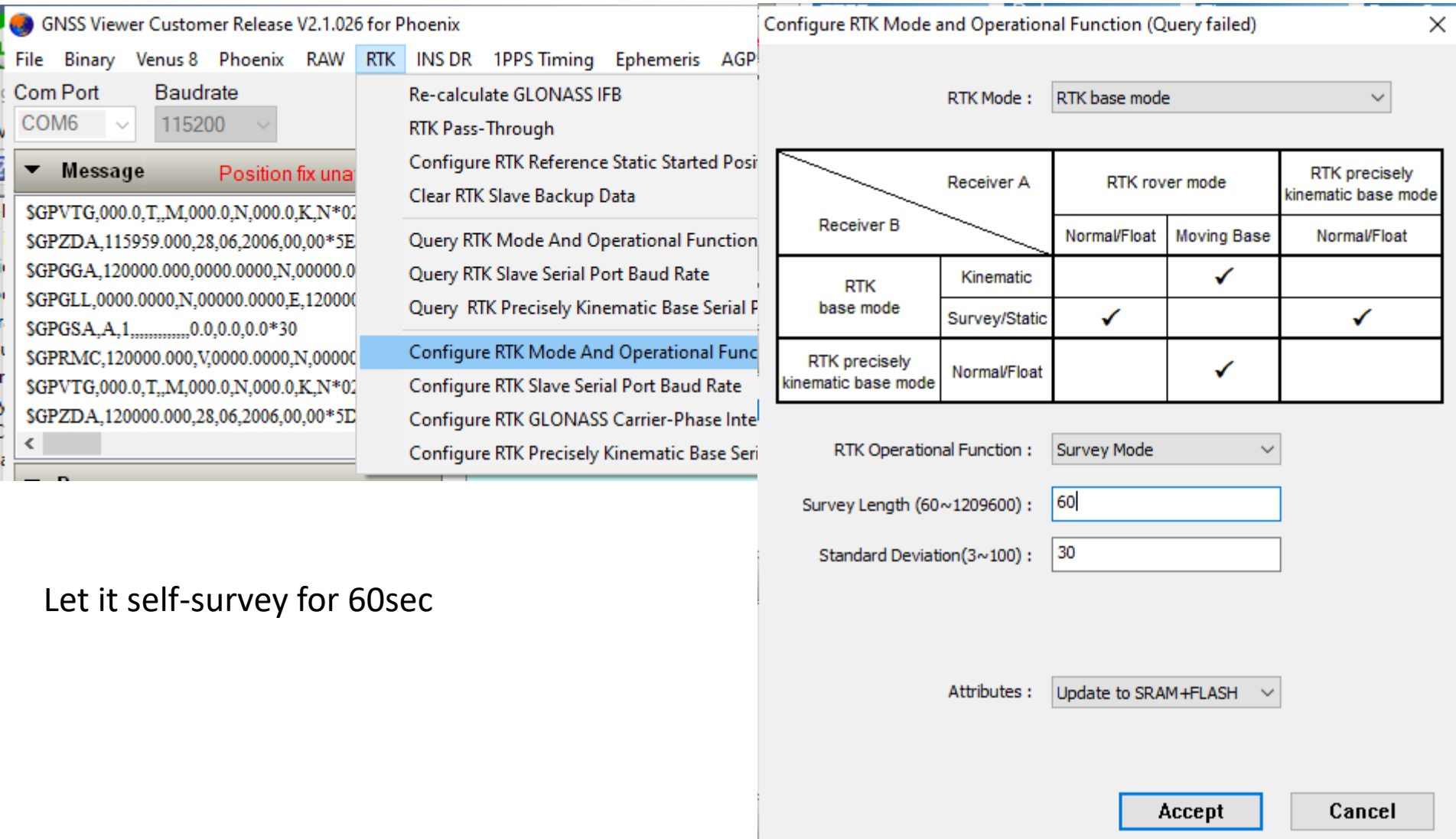
Buttons: Accept, Cancel

Set the antenna position with its known latitude / longitude / ellipsoidal height

ellipsoidal height is defined as
“mean sea level altitude + geoidal separation”

Setup EVB as RTK Base (2/3)

Setup RTK base position: if base antenna position is not known



The screenshot shows the GNSS Viewer Customer Release V2.1.026 for Phoenix interface. The 'RTK' menu is open, and the 'Configure RTK Mode And Operational Function' option is selected. The dialog box for this option is displayed, showing the RTK Mode set to 'RTK base mode' and the RTK Operational Function set to 'Survey Mode'. The Survey Length is set to 60 and the Standard Deviation is set to 30. The 'Accept' button is highlighted.

GNSS Viewer Customer Release V2.1.026 for Phoenix

File Binary Venus 8 Phoenix RAW RTK INS DR 1PPS Timing Ephemeris AGP

Com Port: COM6 Baudrate: 115200

Message: Position fix unavailable

SGPVTG,0.00,0.00,T,M,0.00,0.00,N,0.00,0.00,K,N*02

SGPZDA,115959.000,28,06,2006,00,00*5E

SGPGGA,120000.000,0.0000,0.0000,N,00000.0

SGPGLL,0000.0000,N,00000.0000,E,120000

SGPGSA,A,1,,,,,,,,,0.0,0.0,0.0*30

SGPRMC,120000.000,V,0000.0000,N,00000.0

SGPVTG,0.00,0.00,T,M,0.00,0.00,N,0.00,0.00,K,N*02

SGPZDA,120000.000,28,06,2006,00,00*5D

Re-calculate GLONASS IFB

RTK Pass-Through

Configure RTK Reference Static Started Position

Clear RTK Slave Backup Data

Query RTK Mode And Operational Function

Query RTK Slave Serial Port Baud Rate

Query RTK Precisely Kinematic Base Serial Port Baud Rate

Configure RTK Mode And Operational Function

Configure RTK Slave Serial Port Baud Rate

Configure RTK GLONASS Carrier-Phase Interpolation

Configure RTK Precisely Kinematic Base Serial Port Baud Rate

Configure RTK Mode and Operational Function (Query failed)

RTK Mode : RTK base mode

Receiver A \ Receiver B		RTK rover mode		RTK precisely kinematic base mode
		Normal/Float	Moving Base	Normal/Float
RTK base mode	Kinematic		✓	
	Survey/Static	✓		✓
RTK precisely kinematic base mode	Normal/Float		✓	

RTK Operational Function : Survey Mode

Survey Length (60~1209600) : 60

Standard Deviation(3~100) : 30

Attributes : Update to SRAM+FLASH

Accept Cancel

Let it self-survey for 60sec

Setup EVB as Base Station (3/3)

- After it gets position fix and self-surveyed, the base enters into static mode. You should see the blue dot stop drifting around after static mode is entered from scatter plot view, may need to click Set Origin to see it first. Next power on will still start from survey mode, with self-surveyed antenna location at different place unless static mode is chosen and constant position is entered.
- To see what the self-surveyed latitude, longitude, and height values are, from GNSS Viewer RTK → Query RTK Mode And Operational Function
- To use the base at some fixed location, use above steps to self-survey and retrieve surveyed position, then set antenna position by using the static base mode as described in previous page (1/3)

Setup EVB as RTK Rover

Configure the RTK rover to function in normal RTK mode

GNSS Viewer Customer Release V2.1.026 for Phoenix

File Binary Venus 8 Phoenix RAW **RTK** INS DR 1PPS Timing Ephemeris AGPS DataLog Converter

Com Port: COM6 Baudrate: 115200

Message: Position fix unavailable

RTK Menu:

- Re-calculate GLONASS IFB
- RTK Pass-Through
- Configure RTK Reference Static Started Position
- Clear RTK Slave Backup Data
- Query RTK Mode And Operational Function
- Query RTK Slave Serial Port Baud Rate
- Query RTK Precisely Kinematic Base Serial Port Baud Rate
- Configure RTK Mode And Operational Function**
- Configure RTK Slave Serial Port Baud Rate
- Configure RTK GLONASS Carrier-Phase Inter-Frequency
- Configure RTK Precisely Kinematic Base Serial Port Baud Rate

Configure RTK Mode and Operational Function (Query failed)

RTK Mode: RTK rover mode

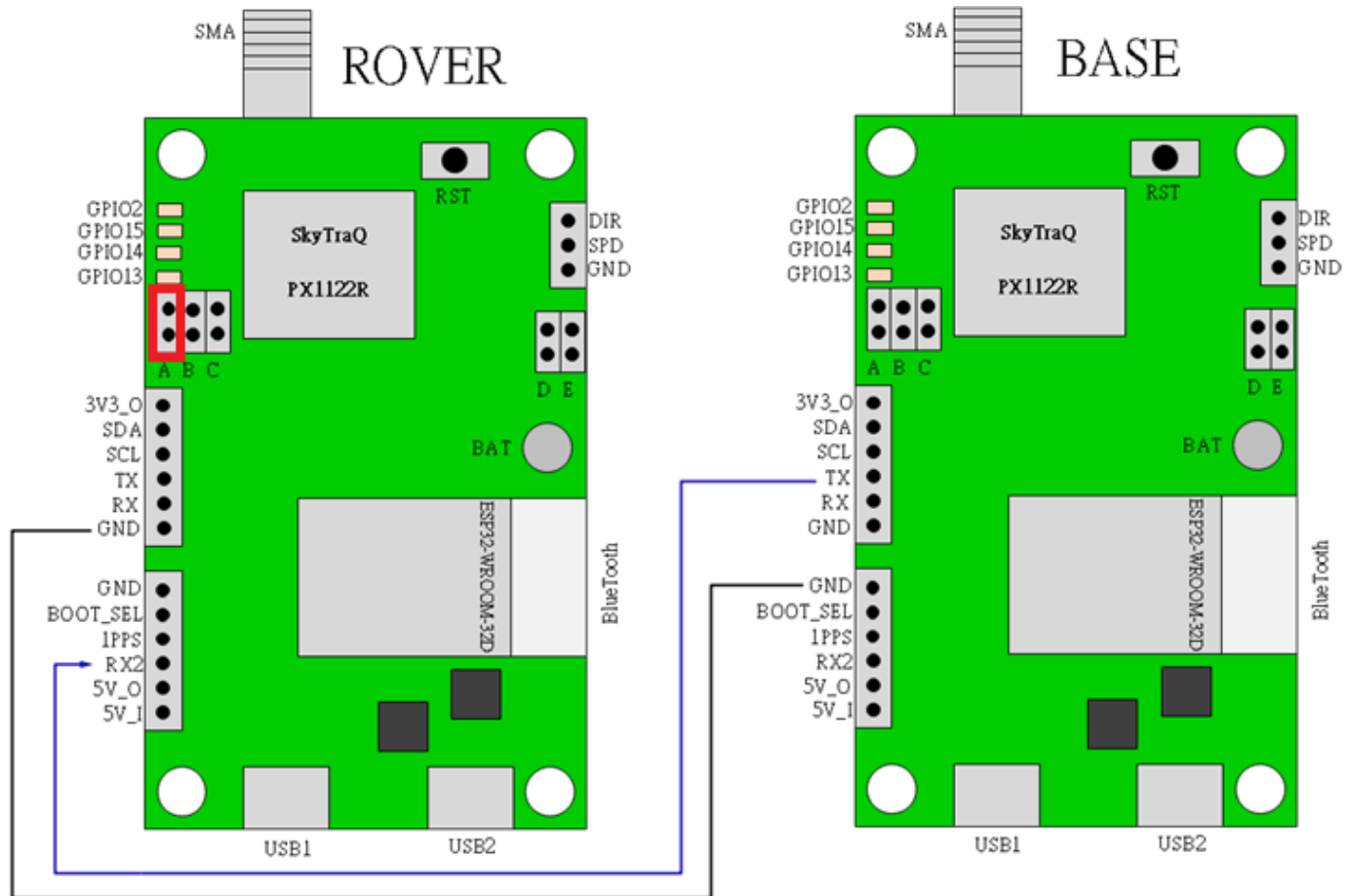
Receiver A \ Receiver B		RTK rover mode		RTK precisely kinematic base mode
		Normal/Float	Moving Base	Normal/Float
RTK base mode	Kinematic		✓	
	Survey/Static	✓		✓
RTK precisely kinematic base mode	Normal/Float		✓	

RTK Operational Function: Normal

Attributes: Update to SRAM+FLASH

Accept Cancel

Very Short Baseline Quick Test (1/2)

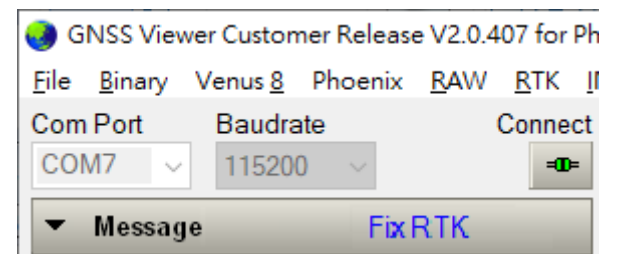
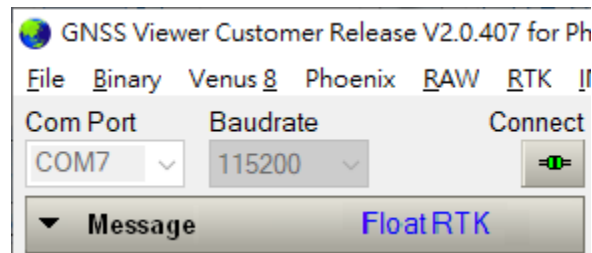
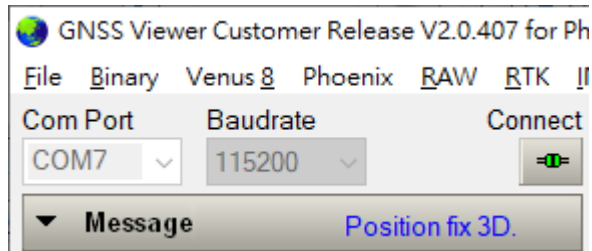


1. Connect ground of base and rover (black wire)
2. Connect base TX to rover RX2 (blue wire)
3. Connect jumper A
4. Plug the USB cables to both USB1 port of base and rover to provide power
5. The baud-rate of base TX MUST be set to 115,200 bps

Very Short Baseline Quick Test (2/2)

- Make sure antenna is under open sky and received signals with CN ratios all over 38dB
- Rover should initially have 3D fix, go through float solution, then later have RTK fix solution

Rover Result on GNSS Viewer



Com Port

COM15

Baudrate

115200

Connect

Message

Fix RTK

\$GBGSV,3,1,09,07,81,101,46,08,75,267,44,10,68,32
\$GBGSV,3,2,09,13,51,236,45,37,39,181,46,30,32,28
\$GBGSV,3,3,09,32,17,160,40,1*4B
\$GBGSV,1,1,04,07,81,101,49,08,75,267,49,10,68,32
\$GNGLL,2447.0954117,N,12100.5264131,E,06331:
\$GNRMC,063315.000,A,2447.0954117,N,12100.5:
\$GNTVG,000.0,T,M,000.0,N,000.0,K,D*16
\$GNZDA,063315.000,13,05,2020,00,00*4D

Response

COORDINATE

WGS84_X	EAST
-2984967.059	-0.005
WGS84_Y	NORTH
4966096.768	0.003
WGS84_Z	UP
2657512.123	-0.001

Command

Hot Start

Warm Start

Cold Start(R)

No Output

NMEA0183

Binary

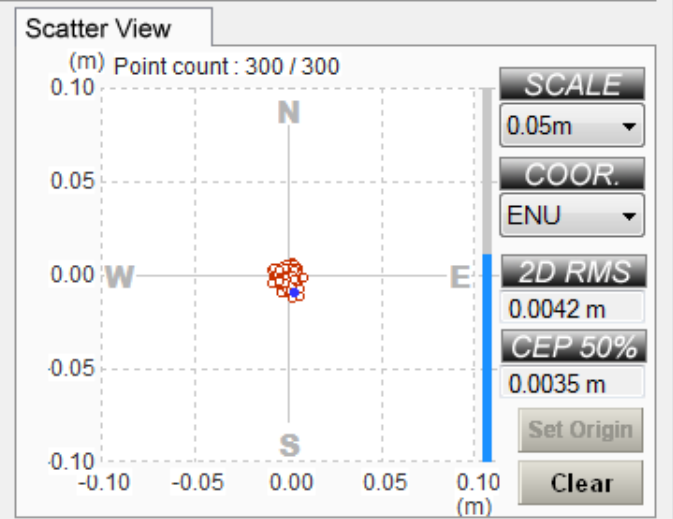
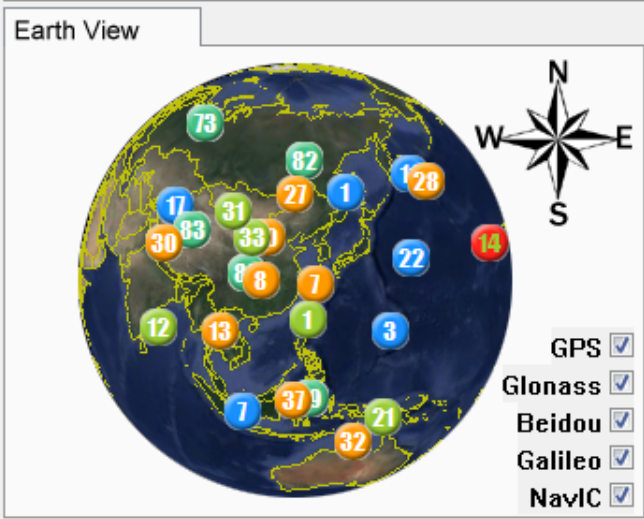
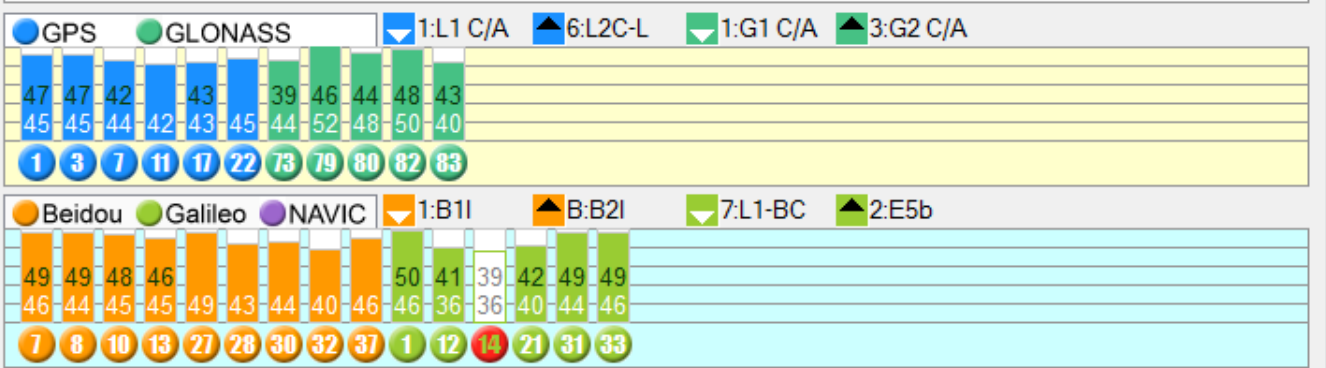
Scan All

Scan Port

Scan Baud

InformationRTK Info.RTK Info2.

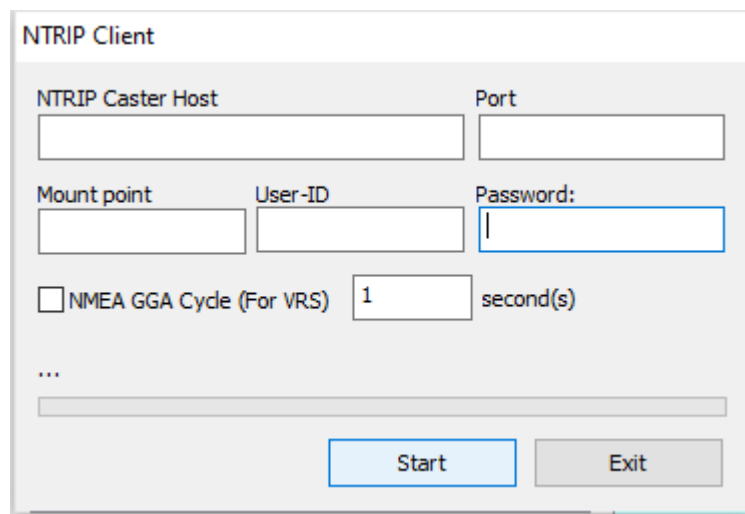
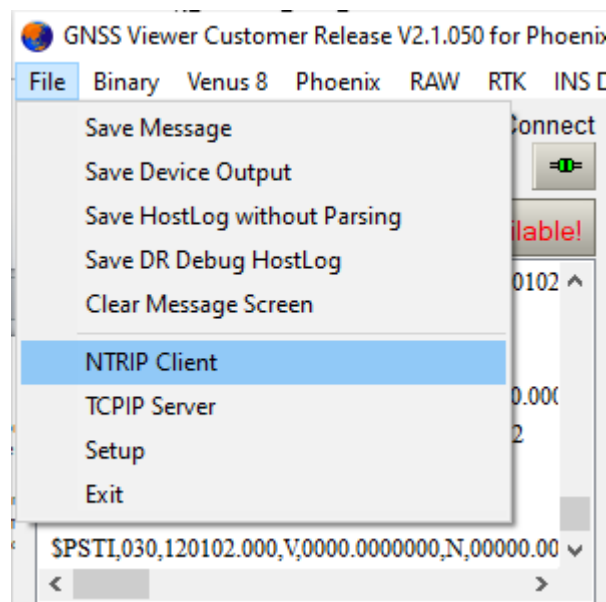
Time	Cycle-Slipped R/B	RTK Age	RTK Ratio	Baseline Length	Baseline Course
06:33:15		1.0	999.9	18.513	22.23
Longitude	Latitude	Ellipsoidal H	East-Projection	North-projection	Up-projection
121°0'31.58489" E	24°47'5.72432" N	116.07	6.947	16.994	2.380



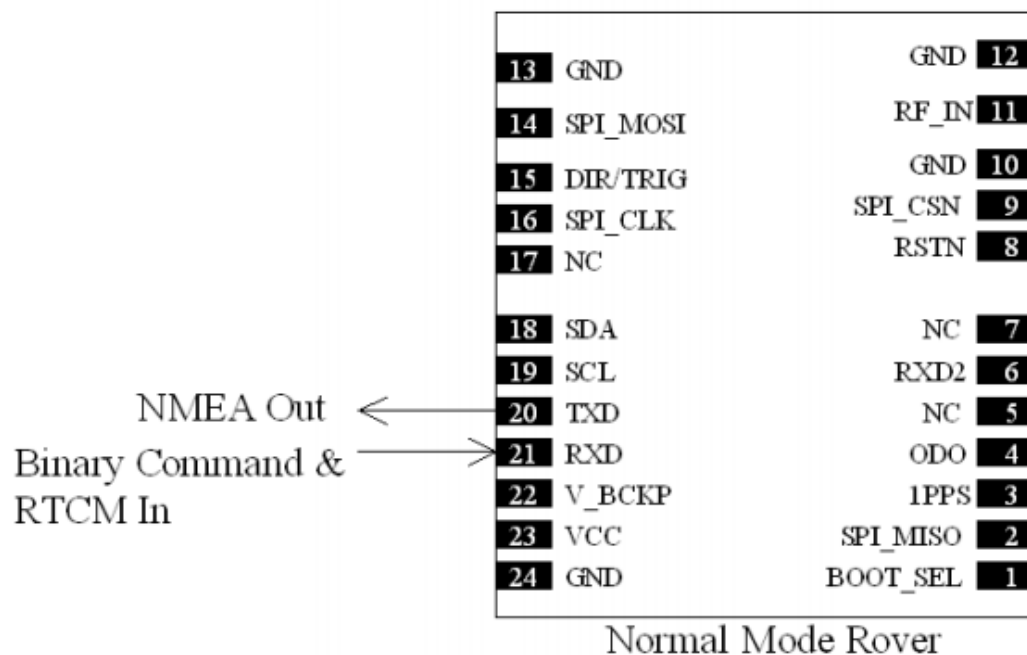
Download

460800

GNSS Viewer 2.1.050 & Later Versions Has Internal NTRIP Client for USB1 Single UART Operation



For Precise Positioning, Rover Mode Configuration 2



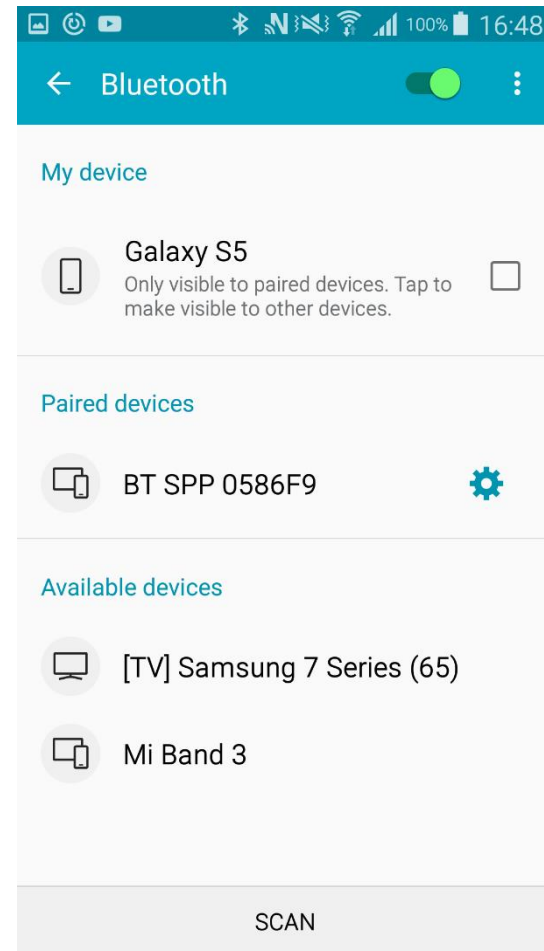
Using Bluetooth with Android Phone (1/8)

- Close jumper C of EVB



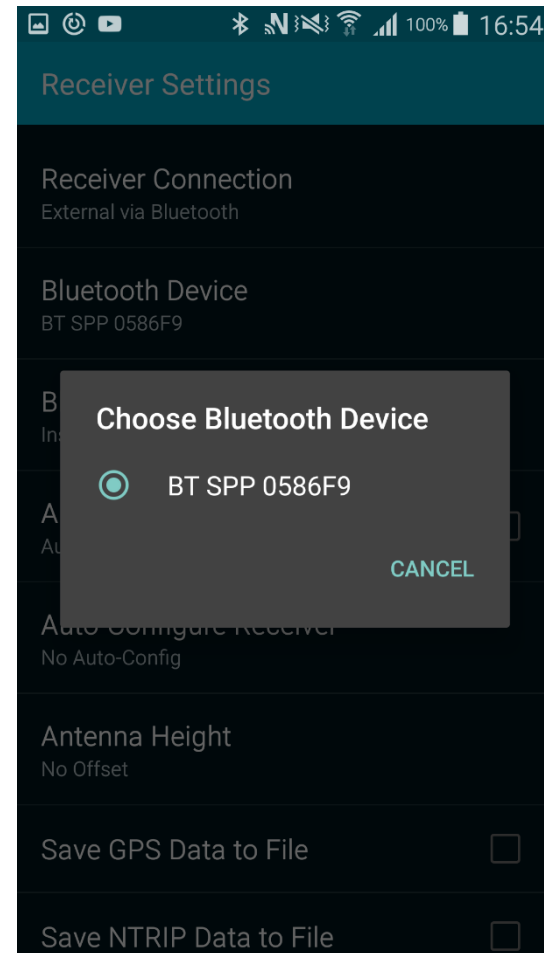
Using Bluetooth with Android Phone (2/8)

- Pair EVB with Android phone



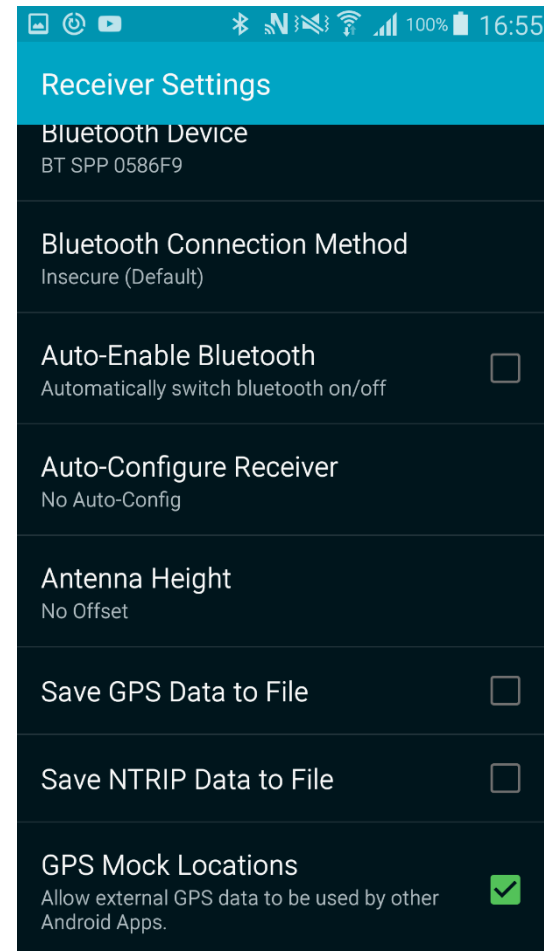
Using Bluetooth with Android Phone (3/8)

- Download Lefebure NTRIP Client app, and set its “Receiver Settings”->”Bluetooth Device” to EVB



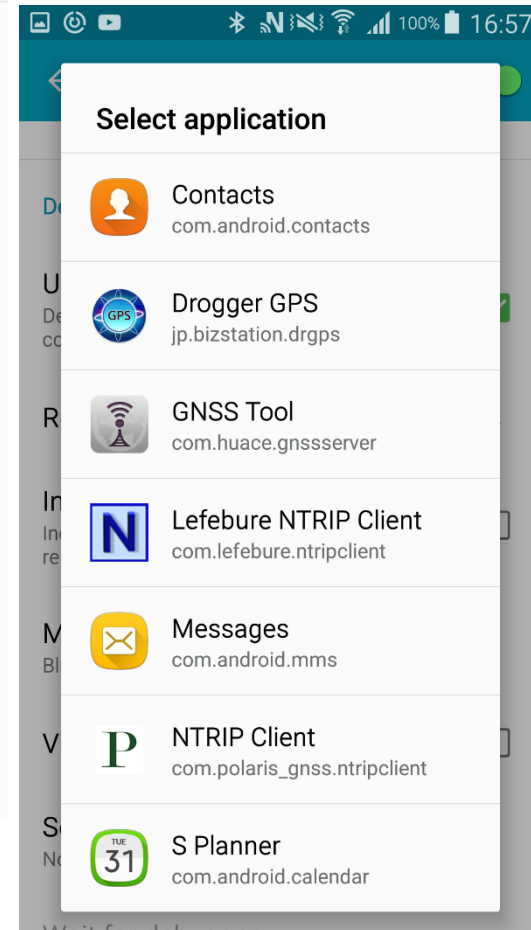
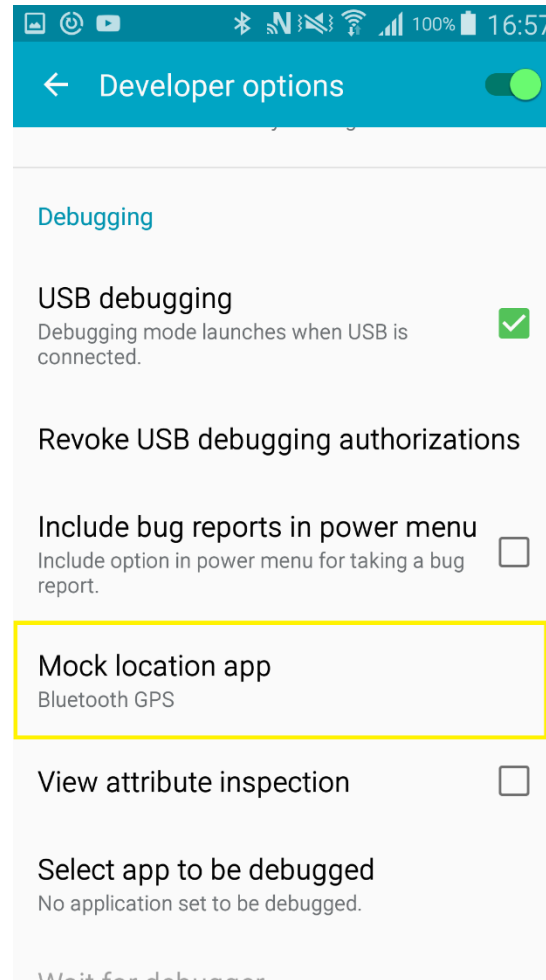
Using Bluetooth with Android Phone (4/8)

- Check “Receiver Settings”->”GPS Mock Locations”



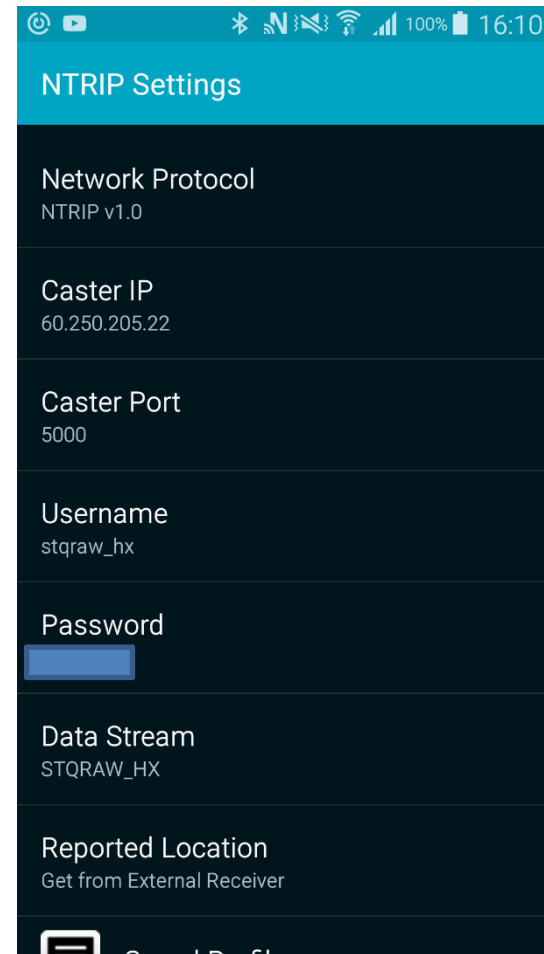
Using Bluetooth with Android Phone (5/8)

- Set System's "Settings" -> "Developer options" -> "Mock location app" to Lefebure NTRIP Client app



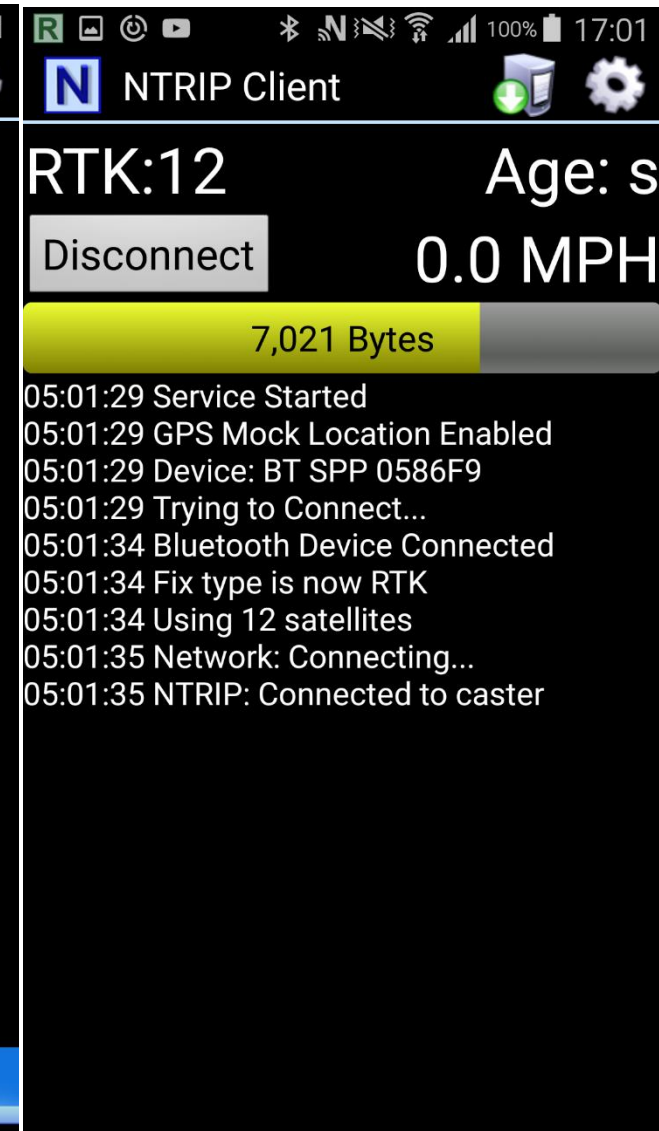
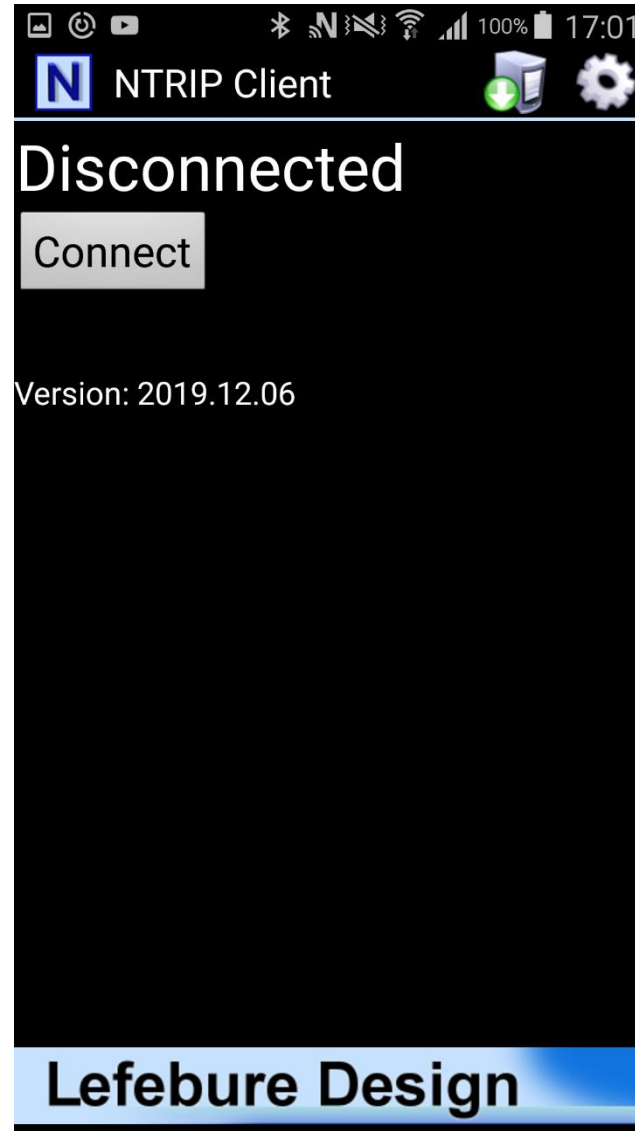
Using Bluetooth with Android Phone (6/8)

- Configure RTK base connecting information at Lefebure's "NTRIP Settings"



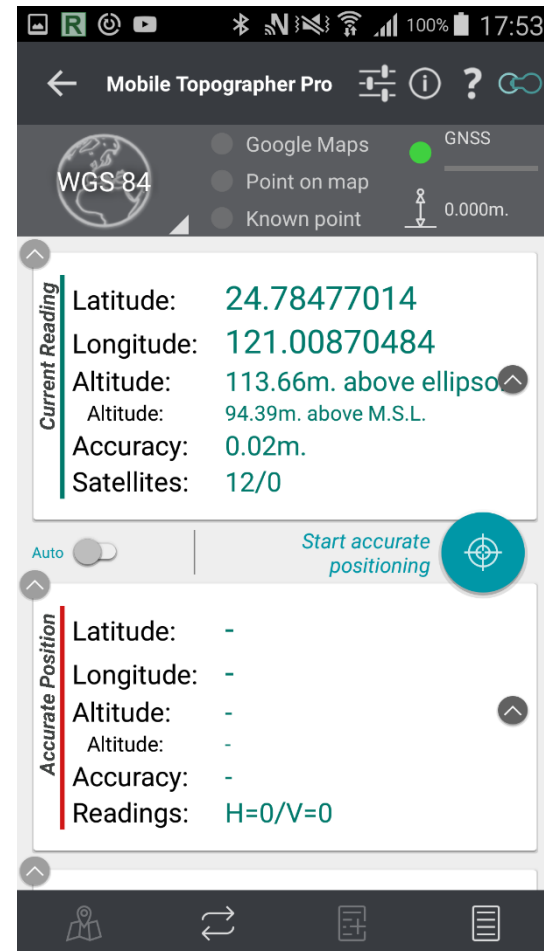
Using Bluetooth with Android Phone (7/8)

- Start Lefebure NTRIP Client by clicking “Connect” button



Using Bluetooth with Android Phone (8/8)

- Now your Android is using RTK coordinates and they can be used by other apps, such as Mobile Topographer.



What To Do Next

- After getting it to work in very short baseline configuration one can extend base and rover distance by using
 - Simple radio link using transmitter and receiver
 - Setup local RTK base to work with RTK rover using Bluetooth/Smartphone NTRIP connection. Follow this guide to setup your RTK base:

[RTK Base Setup for Internet Access](#)

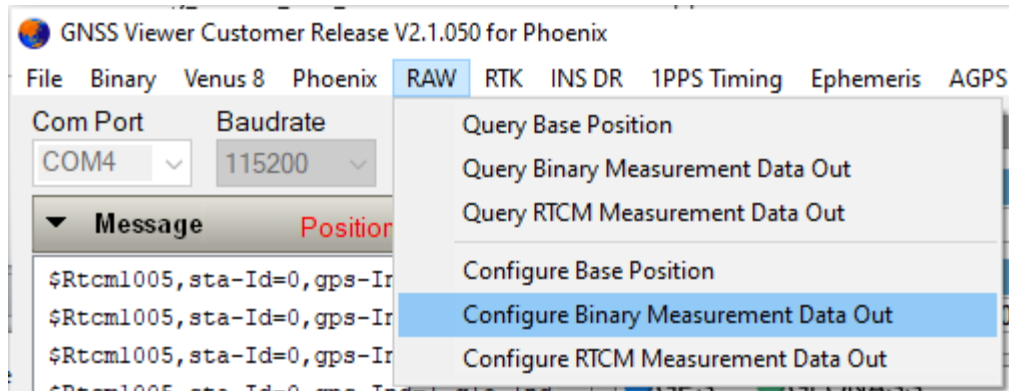
Base Mode MSM4/MSM7 Output Selection

The screenshot displays the GNSS Viewer Customer Release V2.1.050 for Phoenix interface. The 'RAW' menu is open, showing options for querying and configuring base position, binary measurement data, and RTCM measurement data. The 'Configure RTCM Measurement Data Output' option is selected. A secondary window titled 'Configure RTCM Measurement Data Output (Query successfully)' is open, showing the following settings:

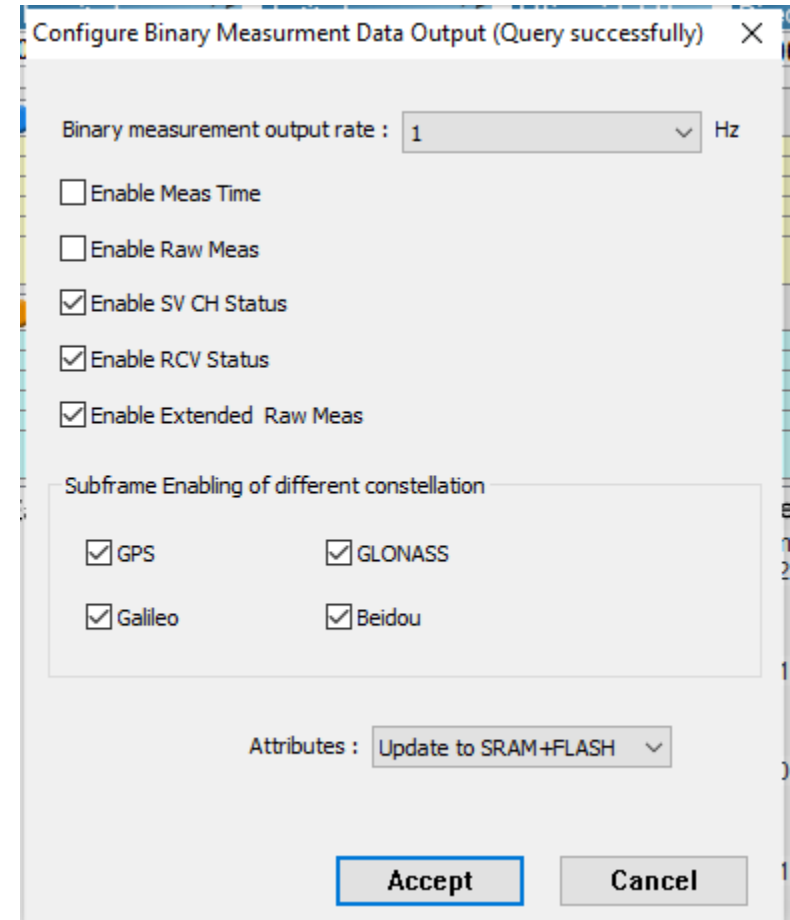
- ☒ Enable RTCM Output
- Output Rate for MSM: 1 Hz
- Output MSM Type: MSM7
- ☒ Stationary RTK Reference Station ARP (Message Type 1005)
- ☒ GPS MSM7 (Message Type 1077)
- ☒ GLONASS MSM7 (Message Type 1087)
- ☒ Galileo MSM7 (Message Type 1097)
- ☒ SBAS MSM7 (Message Type 1107)
- ☒ QZSS MSM7 (Message Type 1117)
- ☒ BDS MSM7 (Message Type 1127)
- ☐ GPS Ephemeris(Msg Type 1019) Interval (1 ~ 255): 0 second(s)
- ☐ GLONASS Ephemeris(Msg Type 1020) Interval (1 ~ 255): 0 second(s)
- ☐ Beidou Ephemeris(Msg Type 1042) Interval (1 ~ 255): 0 second(s)
- ☐ Galileo Ephemeris(Msg Type 1046) Interval (1 ~ 255): 0 second(s)
- Attributes: Update to SRAM+FLASH

Default base mode output MSM4 message. It has lowest data bandwidth requirement. 9600 baud rate is sufficient.

Base Mode Binary Output Configuration



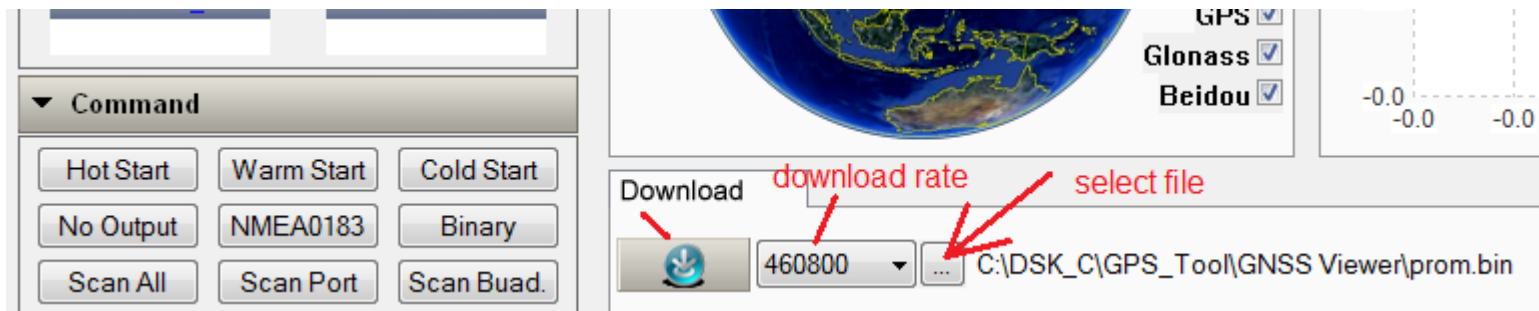
This binary output mode normally not used due to excessive data bandwidth needed, minimum of 115200 baud rate required.



Firmware Update

To update firmware to PX1122R EVB:

- 1. Unzip the firmware file
- 2. Connect EVB USB1 to a Windows PC running GNSS Viewer. Connect GNSS Viewer to the EVB.
- 3. Select the unzipped .bin file using the small button on right of the Download button.
- 4. Click the Download button, first boot-loader will be downloaded, and then main firmware will be downloaded. After downloading is done, the pop-up window will close and module will reboot.



Radios That Some Customers Use with PX1125R

- XBee-PRO 900HP
<https://www.digi.com/products/embedded-systems/digi-xbee/rf-modules/sub-1-ghz-modules/xbee-pro-900hp>
- XTend 900 MHz
<https://www.digi.com/products/embedded-systems/digi-xbee/rf-modules/sub-1-ghz-modules/xtend-module>
- RFD900+
https://www.alibaba.com/product-detail/RFD900-Plus-915MHz-Ultra-45km-Long_60812568500.html
- xbee s3b
https://www.alibaba.com/trade/search?fsb=y&IndexArea=product_en&CatId=&SearchText=xbee+s3b
- e61 433t30d
https://www.alibaba.com/product-detail/Ebyte-Hot-sale-high-speed-wireless_60654442645.html